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UNIT - I

Introduction to Civil Engineering and Construction Materials

BASICS OF CIVIL ENGINEERING

- Meaning of the word 'civil' is about cities and citizens.
- Civil engineering aims at providing basic services, facilities amentities and needs of the people in urban as well as rural areas for efficient and economical distribution of the available resources in nature.
- It is the oldest branch of engineering which involves surveys, planning, designing, construction and maintenence
- Civil engineering plays a vital role in all the other branches of engineering also. Out of every fundamental human needs (food, shelter, clothing) it directly or indirectly deals with power supply, irrigation systems, environmental engineering etc.
- Advanced instruments and techniques such as digital measurements, Global positioning system (G.P.S), Geographical Information system (G.I.S), Satellite Remote sensing are used for least errors, fast work and efficient resources management.

STRUCTURAL ENGINEERING

→ It is a branch of civil engineering that includes safe and economical design of structures and structural members as well as connections such as rivets, bolts, keys, welds etc.

→ For a given loading, suitable ~~a~~ cross section of members (beams, columns etc) can be determined.

→ Structural engineering further includes engineering mechanics, strength of materials, theory of structures design of steel and reinforced cement concrete (RCC) structures etc.

→ For the given loading, conventional materials as well as alternate / modern materials are used to get minimum possible size of structural members with adequate factor of safety.

APPLICATIONS

→ Design and construction of structural members as well as structures, connectors with adequate factory of safety and economy.

→ Design of superstructure and substructure (foundation) of a building or factory shed etc

→ Design of steel components and structures including water tanks.

→ Design of concrete members, rigid pavements, bridges etc

GEOTECHNICAL ENGINEERING

→ It is a civil engineering discipline that deals with study of soil properties and engineering behaviour under the action of particular loads and moisture content.

→ Physical properties of soil has great impact on stability and safety of structures.

→ Particle size, chemical composition, moisture content and swelling - shrinking of soil which must be considered for design of foundations, roads, buildings etc.

→ This discipline is very close to structural engineering

APPLICATIONS:

→ Determination of important physical and engineering properties of soil lying underneath.

→ Effective, effecient and economical type of foundation for load transfer on wider areas.

→ Estimation of bearing capacity (B.C) of soil and improving the B.C

Design of retaining walls, earthen dams / embankment.

- Improving the stability to control landslides
- Improvement in foundation design, type and construction for specific requirements like atomic power plant, airports, harbours / ports
- Study of sub-soil layers and soil profile for preparing of soil maps.
- Advanced studies and applications such as estimation of depth of water table, ground water, seismic activities, fault movements at different depths below ground level.
- Design and construction (as well as maintenance) of transportation routes and allied structures such as bridge, tunnel, shafts etc
- Water seepage analysis for dams

WATER RESOURCES AND IRRIGATION ENGINEERING:

- The civil engineering that deals with tapping or storage of water and supplying water either for crop cultivation or for drinking and other domestic / industrial uses.
- Two important basic human needs water and food taken care of by irrigation engineering directly and indirectly.
- Water is an important and scarce natural resource is in the form of surface water and sub-surface water which is stored in dams and reservoirs or tapped from wells.
- An irrigation project may be multipurpose ie
- for water supply as well as for hydropower generation
- This engineering includes estimation of quality and quantity of water available, its storage and distribution through open canals, or pipe-networks either under gravity or by pumping.

APPLICATIONS

- Estimation of quality and quantity of surface and sub-surface water available at different places
- Water supply schemes for cropping, drinking and other purposes
- Design and construction of reservoir capacity, and dams.
- Design and laying canals, pipes, pumps and allied structures such as spillways, gates etc
- Flood control arrangements, diversion works, etc
- Avoiding water logged areas and salination of soil and also control soil erosion
- Estimation of water requirement for cropping, so that most effecient and economical use of water is made.
- Providing non-conventional effective techniques such as drip irrigation, sprinkler irrigation etc.

HYDRAULICS

- Hydraulics is an applied science that studies pipe flows and open channel flows, & dams and irrigation, hydropower, hydrodynamic machines, fluid control circuitry, etc.
- The below figure gives allied branches and general classification of 'mechanics' oriented basic fields.

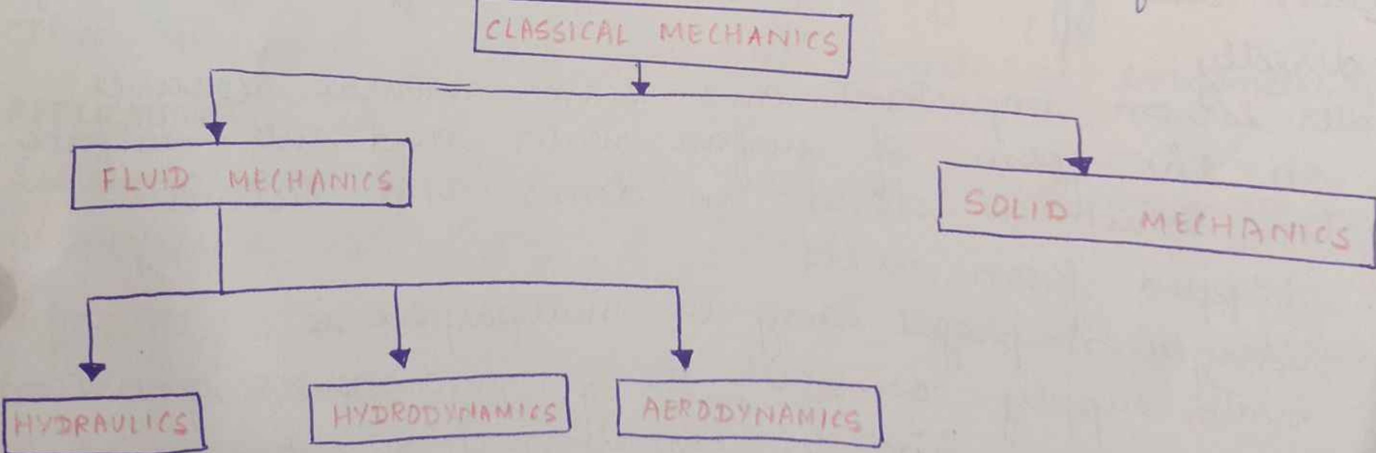


Fig: Fields related to 'Mechanics'

[...]ICATIONS:

- Determination of fluid properties such as density, viscosity, vapour pressure etc
- Control and monitoring of flow of oils, gas/air, water etc
- Design of valves, gates
- Design of conveying systems for fluids
- Design of rotodynamic machines (pumps, turbines etc)
- Design of hydropower system
- Storm water management
- Design of streamlined bodies, boats, ships etc
- Determination and minimizing energy losses
- Measurement of fluid flow parameters (velocity, pressure etc)

TRANSPORTATION ENGINEERING:

- This is the discipline that deals with study of present transporting systems and their improvement for safe, economical and effecient, conveyance of materials or goods or finished products as well as human beings and animals.
- It includes design construction and management of roads, railways, navigation and air routes. Allied constructions such as funnels; bridges, culverts acqueducts
- Traffic management and parking faclities
- Design and construction of different types of roads.
- Design and provision of curves and allied structures such as bridg, culvert, tunnel, ghat-roads
- Use modern techniques of management to ensure rapid transportation of people and goods with sufficient convince, comfort, economy and safety
- Avoid heavy traffic through cities / villages by providing bye-pass, diversion roads, expressway.
- Help the economic growth of regions and country through fast transportation system

→ Provide durable and strong as well as safe mode of transport and repair / maintenance with least possible delays and inconvience

MODES OF TRANSPORTATION

→ Roadways:
  ⇒ National highways
  ⇒ State highways
  ⇒ Major district roads
  ⇒ Minor district roads
  ⇒ Village roads

→ Types of roads:
- Bituminous road
- Water bound mechadam Roads
- Concrete road
- Eartheen roads

→ Railways → Laying of new tracks, allignment of the track, crossing, sharp curve to be avoided. Uneven land to be evened by filling the gap. Super elevation near crossing to overcome centrifugal force

→ Airways: Airport design, construction of runways ⇒ should be designed by taking into consideration of load on the road. - specially impact load.

→ Waterway: Construction and maintenance of harbour.

ENVIRONMENTAL ENGINEERING

Deals with study of the natural environment, inter-relation between biotic and abiotic factors, safety of people against different types of pollution and treatment disposal of wastes

→ Measurement of pollutants as regards air, water, radioactive, and noise pollution.

water treatment for supplying potable water.

Waste water treatment plant - design, construction and repair maintenance

Research and development for recycling the mass of energy from wastes,

→ Determining / fixing standards for effluents

→ monitoring, control, prevention of different types of pollution.

→ Environmental Impact Assessment / analysis for factories and industries

→ Conservation and preservation of natural resource and Environment.

→ Design, construction and maintenance of water filters and water supply schemes.

→ Testing and reporting of pollutants such as industrial effluents for enforcement of relevant laws

CONSTRUCTION PLANNING AND MANAGEMENT

Deals with study of different types of constructions of structures with requisite economy, efficiency and factor of safety

→ It also includes excavation, construction of foundation and footing, concreting, finishing, bridges, tunnels underground and underwater construction with different techniques and modern methods and construction machinery

→ Construction management: Project management and systematically completion of a construction work ie systems approach, considering special requirements for the construction.

→ Maximize and minimize, optimization of material, labour money and time required for construction.

→ use of modern and effective methods (bar chats) and construction machines, use of smart and alternative/ composite materials for construction

→ Scheduling and phasing of works for managing the operations and stages involved in the construction.
→ Achieving good quality of work with economy, efficiency and factor of safety as guiding principles.
→ Ensuring expected strength, durability and workmanship of the constructions.
→ Following relevant acts, rules and regulations, bye laws and specifications regarding construction work and materials of construction.

BASIC MATERIALS OF CONSTRUCTION

• BRICKS

Made of clay or earth containing about 30% alumina 40% of silica and other constituents like magnesia, ferric oxide, calcium carbonate about 10%.

Excess alumina → Shrinkage and warping effects are observed on drying or burning.

Excess silica → Brick brittle.

→ Manufactured locally and gives good thermal insulation when used for masonary construction. Easily available at cheaper rates.

Requirements of good brick

→ Uniformly and thoroughly burnt, with uniform colour and straight edges.
→ Should be hard, giving metallic ringing sound when struck together. No scratch is made on its surface of by finger nail.
→ Free from cracks, flaws, lumps and holes.
→ minimum compressive strength of 3.5 N/mm^2
→ Should not absorb water more than 20% of its dry weight.
→ Should not break into pieces when dropped freely from a height of about 1m on the ground.

→ Class I Brick: Exterior masonary walls, short columns and arches,
→ Class II ⇒ Brick used for internal walls, compound walls.
→ Class III ⇒ Brick Bat lime concrete (B.B.L.C) small foundations, passings, brick flooring.
→ Coloured / sand-lime bricks are used for architectural, ornamental works.
→ Hollow bricks are used for multistoreyed buildings, sound proofing, heat insulation walls.

ADVANTAGES

→ Light weight compared to stones.
→ Better strength, better fire, sound and heat resistance easy to work.
→ Locally manufactured, easily available at cheaper rates
→ Uniform in size helps wall construction of uniform thickness.
→ Special or trained labour is not required for ordinary works.

• CEMENT

→ It is an artificial binding material manufactured by burning the mixture of calcarious (containing lime - calcium), silicious (containing silica) and argillaceous (contain alumina) materials in specific proportion at a very high temperature.
→ Commonly used greyish coloured cement is known as ordinary portland cement (O.P.C)

USES:

→ Wall-construction for buildings, retaining walls, dams, bridges, pavements etc.
→ Plastering and pointing of wall and flooring
→ Interior / exterior decorative work.

- manufacturing of building blocks, precast components
- Cement grouting for 'sealing' and cracks / gaps to prevent leakage or seepage of water
- When water is mixed with cement, chemical action takes place, which is of exothermic type. Hence heat is generated, known as heat of hydration or heat of evolution. If the heat is not properly dissipated, cracks are formed. This reduces strength, stability and durability

QUALITY OF CEMENTS

- I.S 269 – O.P.C, rapid hardening and low heat cement
- I.S 455 – for blast furnance cement
- I.S 1489 – for pozzolona cement.

Recent types

1. 33 grade – minimum compressive strength $33 N/mm^2$
2. 43 grade – minimum compressive strength $43 N/mm^2$
3. 53 grade – minimum compressive strength $53 N/mm^2$

- It is a primary element with coarse aggregate < 1.25mm

MORTAR

$\boxed{\text{MORTAR} = \text{CEMENT} + \text{WATER} + \text{FINE AGGREGATE}}$

- Used for joining bricks
- Forms strong joints
- Ratio : 1 : 1.5

cement   fine aggregate

- CONCRETE

$\boxed{\text{CONCRETE} = \text{CEMENT} + \text{WATER} + \text{COARSE AGGREGATE} + \text{FINE AGGREGATE}}$

[P]lain and Reinforced concretes

If reinforcement in the form of metal bars in the moulds is used before placing the concrete mix, it is called Reinforced cement concrete (R.C.C)

USES OF R.C.C

→ Extensively used for retaining walls, concrete roads, bunkers, water tanks, multistoryed buildings, buildings foundations, framed structures.

Concrete + steel ⇒ RCC

⇓ ⇓

Compression ↓

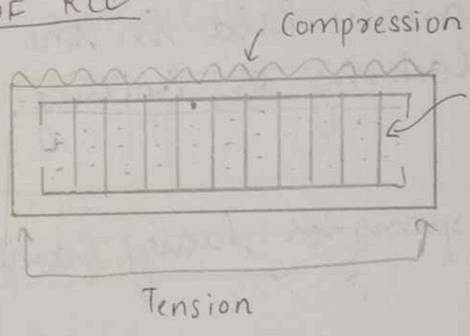
comp ↑ tensile ↑

tensile ↓

→ RCC ⇒ has both compressive and tensile stress's

→ RCC = cement + fine and coarse aggregate + water and steel.

BEHAVIOUR OF RCC



Stirraps for holding longitudinal reinforcement in position.

P.C.C

→ If reinforcement bars are not used, it is called plain cement concrete (P.C.C)

→ used for steps, flooring, bed concrete

→ It has high compressive strength only, almost no tensile strength.

Pre-stressed concrete (PSC)

(IS . 1343 - 2012)

→ Principles: Concrete once stressed, behaves as one homogeneous, isotropic and elastic material.

→ PSC structural behaviour is same as RCC

→ For very long span, prestressed steel acts as suspension cable and concrete around it as stiffening girders

Types

• POST-TENSIONED (PSC)

(M30 concrete is used)

→ Induced after setting and hardening of concrete mass

→ The wires are in a bundle inside a sheathing

• PRE-TENSIONED (PSC)

(M40 concrete is used)

→ Inducing the prestress before setting and hardening of concrete

→ Once stressed, the wires can be held in the stressed position by anchoring, locking or bolting.

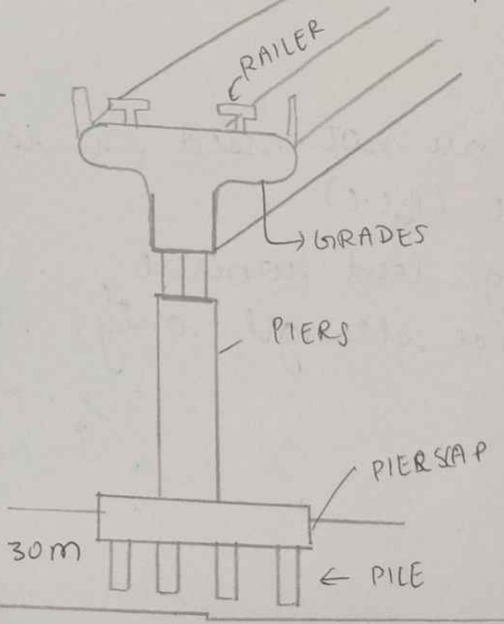
USES OF POST-TENSIONED (PSC)

→ For big works with long spans or heavy loads such as bridges, concrete Tanks

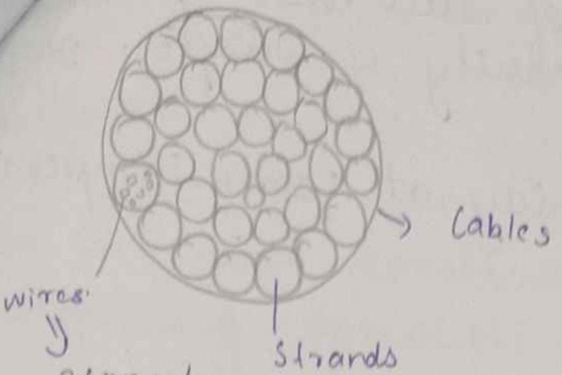
USES OF PRE-TENSIONED (PSC)

→ Used for small works such as short beams, railway sleepers.

SAMPLE OF PSC



M30, M40 concrete
use of high strength concrete and high strength steel



$2600\ N/mm^2$ (Tensile strength)

Diameter of wire = 1.5 – 6mm

Grades of Concrete

M10 —— compressive strength

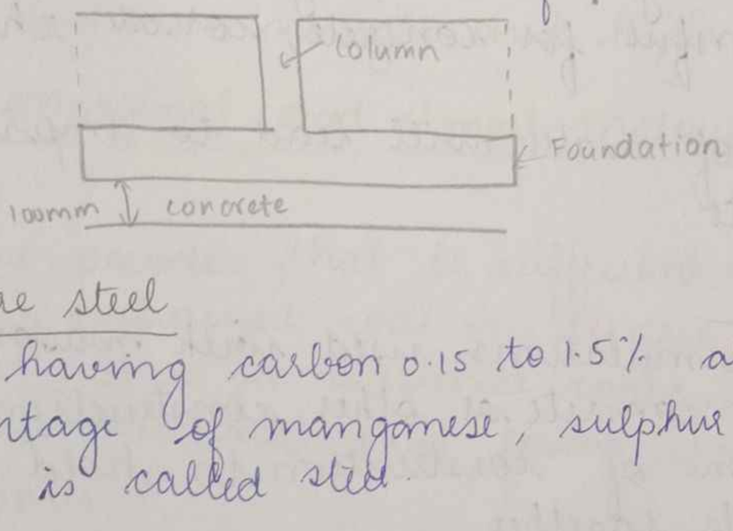
M15, M20, M25, M30 } used for residential use (for apartments)

M60, M80 } high strength concrete (for dams)

cube (size of concrete cube) = 150 X 150 X 150mm

M30, M25 => used for foundation

M15, M20 => constructions of beams + columns + slab



Structure steel

-> Iron having carbon 0.15 to 1.5% along with small percentage of manganese, sulphur, phosphorous and silica is called steel

Grade of steel

Fe $\boxed{250}$ → indicates tensile strength of steel

Fe 415

Fe 500

Fe 550

Fe 600

Structural steel is a category of steel used for make construction materials in a variety of shapes. It is also known as mild steel.

- mild steel has good workability at room temperature or high temperature
- Good elasticity and ductility.
- can be hardened by tempering.
- Galvanisation can be done
- They have good tensile, compressive as well as shear strength
- They can be used for preparing screws, steel girders => sanitary fittings, pipes, tubes, water tanks, sheets, rails, cables, roofing sheets, door / window frames

Water

It is an important element which helps cement in hydration. Proper quantity of water, cement ratio should be maintained.

- Used in preparation of concrete, RCC, cement.
- It helps in strengthening the building constructed.
- Hard water is harmful for concrete, corrode the reinforcement.
- Excessive amount of water will lead to improper bonding with concrete.

Construction Chemicals

- They are chemical formulations used with masonary materials, cement, concrete or other construction materials at the time of construction to hold the construction materials together
- Some of them are: Protective coating, decorative coating, concrete hardeners, concrete curing, mould releasing agents, Polymer bonding agent, ready mix plaster, polymer modified mortar, waterproofing chemicals

CARBON COMPOSITES: They use the strength and modulus of carbon fibres to reinforce a carbon matrix to resist the rigors of extreme environments.

These carbon fibre materials can be used in different stages such as construction stage as a part of reinforced concrete used to resist the tensile forces and stresses.
used in post construction stage in repairing and strengthening operation of structural elements.

Plastics in construction

- It is a perfect building material
- It is cheap, available and easy to mould.
- Easy to convert into building materials.
- It is durable, waterproof, insulating making it suitable for building in many different types of climate.
- It is used for construction mainly for seals, profiles pipes, cables, floor coverings and insulation
- They are even used for aesthetic purpose.
- It can be moulded into any shape, size like pipes, cables, panels, sheets, coverings can be manufactured.
- It is light in weight and high tensile strength

New, marginal and Smart materials

FLYASH

Fine powder that is suffecient × byproduct of burning pulverized coal in electric generation power plants. It is a substance containing aluminous and siliceous material that forms cement in presence of water.

$$\boxed{\text{FLYASH} = \text{FLYASH} + \text{LIME} + \text{WATER}}$$

- Similar to PCC cement
- It replaces PCC and is cost effecient than PCC
- Suitable as a prime material in blended cement, mosaic tiles and hollow blocks.
- when used in concrete mixes, flyash improves the strength and segregation of the concrete mixes;

flyash improves the strength and makes it easier to pump.
→ The particles are spherical in shape which helps to reduce friction between aggregates thus increasing the workability of concrete

New age concrete

Concrete created ~~used~~ using flyash cement, which has high tensile strength and is cost efficient.
→ This concrete has high workability.
→ It is easy to pump

Recycling of materials:

Various types of recyclable materials are currently used in civil engineering
→ These include fire rubber, fly and bottom ash, blast-furnance slag, steel slag, cement kiln dust, silica fume, crushed glass.
→ Wood, earthen materials, bricks, concrete can be recycled.
→ Recycling waste reduces disposal costs and carbon emission.
→ It also helps you comply with environment legislation and restrictions on what can be sent to landfill.
→ It saves energy and decreaces the consumption of natural resources to produce new materials

Applications of smart and sustainable materials in civil engineering constructions

→ Smart materials are used in the designs of smart buildings.
→ They are used for vibration control, noise mitigation safety and performance
→ Smart materials are used in smart building for environmental control, structural health monitoring.

Smart materials play role in the construction of road pavements as a traffic - sensing recorder and also melts ice on highway and air fields during snowfall

Smart materials are also used to monitor the civil engineering structure to evaluate their duralitity

→ They are used to monitor the integrity of bridges, dams.

→ Sustainable material like Timber has lowest environmental impact on its production and life cycle. It is sustainable and recyclable.

→ Isolation renewable materials : These are completety recyclable and compostable such as cellulose, they do not generate wastes and used in construction and aesthtic purpose of the building.

→ Wood is the reusable and renewable even bamboo wood, beeswax they are all used in aesthtic purpose and wood for flooring

→ Using these materials will be environmentally friendly, thereby reducing emissions, illness and energy consumption.

SOFTWARE'S USED

→ ETABS, SAP, STAAD-PRO (for analysis of building structure)

→ These softwares help civil engineering throught design and construction processes

→ Auto CAD civil 3D (Drafting software)

→ HEC- HMS (stimulate the hydrologic process of watershed systems)

→ Storm CAD (Analysis program for drainage)

→ Autodesk Revit (for structural engineer)

→ 3DS max (3D modeling)

→ Revit for BIM (Building information modeling)